

NLU course project

Matteo Grisenti (257855)

University of Trento

matteo.grisenti@studenti.unitn.it

1. Introduction

This project began with a basic RNN model, initially experimenting only with the learning rate. As the work progressed, other hyperparameters were gradually adjusted to better understand their individual effects on performance. This incremental tuning approach helped maintain control over the training process, reducing complexity and improving interpretability. The main hyperparameters explored in this study include: learning rate (LR), batch size, dropout rate (DO), hidden and embedding dimensions, and the number of layers (#L).

The experimental process was divided into two phases: a training-validation for hyperparameter tuning, followed by a final testing of the best-performing model on a separate test set.

2. Implementation Details

The implementation was based on the initial code provided in Lab 4, which was gradually modified to align with the objectives of this project.

Each model was trained for a maximum of 100 epochs, with evaluations conducted at the end of every epoch. Early stopping was implemented if no improvement was observed over three consecutive evaluations. The model with the best validation performance was saved, along with its corresponding evaluation metrics. At the end of each training run, plots were also generated to visualize the progression of both the loss and perplexity (PPL) over time.

Validation metrics were recorded in the file `dev.csv`, and include 95% confidence intervals (CI) for the perplexity (PPL) scores. After training, the best model was evaluated on the test set, with the final results saved in `test.csv`.

3. Results

3.1. PART 1A

To establish a baseline, a simple RNN model was trained using learning rates (LR) of 5, 1, and 0.1.

RNN Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	5	128	200	300	/	232.33
1	1	128	200	300	/	164.44
1	0.1	128	200	300	/	195.96

Table 1: Results RNN Networks

LSTM: We then evaluated the LSTM architecture with learning rates of 0.5, 1, and 2, and batch sizes of 16, 32, 64, and 128. Results show that the LSTM consistently outperforms the RNN in terms of perplexity. Additionally, smaller batch sizes (16 and 32) led to better performance, suggesting that larger batches (64 and 128) may hinder the model's ability to learn fine-grained patterns.

LSTM Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	0.5	16	200	300	/	146.94
1	0.5	32	200	300	/	149.50
1	0.5	64	200	300	/	155.47
1	0.5	128	200	300	/	155.39
1	1	16	200	300	/	145.96
1	1	32	200	300	/	146.62
1	1	64	200	300	/	150.28
1	1	128	200	300	/	154.23
1	2	16	200	300	/	145.42
1	2	32	200	300	/	144.84
1	2	64	200	300	/	148.38
1	2	128	200	300	/	150.38

Table 2: Results LSTM Networks

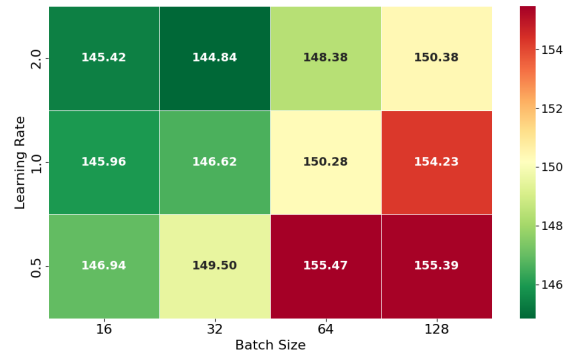


Figure 1: PPL Heatmap of the LSTM Network models

DROPOUT: Starting from the best-performing LSTM configuration (LR=2, batch size=32), we added two Dropout layers—one after the embedding layer and one after the LSTM. This resulted in a notable improvement in performance and training stability. We experimented with dropout rates of 0.1, 0.2, and 0.5. While 0.1 and 0.2 produced similar results, a dropout rate of 0.5 caused underfitting, though it still outperformed the vanilla LSTM.

LSTM (DropOut) Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	2	32	200	300	0.1	129.26
1	2	32	200	300	0.2	126.47
1	2	32	200	300	0.5	138.26

Table 3: Results LSTM Networks with 2 Dropout Layers

ADAMW: High learning rates (e.g., LR=1) were unsuitable, causing unstable training. Experiments with a fixed batch size of 32 and dropout rate of 0.2 across learning rates of 0.1, 0.01, 0.001, and 0.0001 showed that smaller values improved performance. Although LR=0.0001 yielded the lowest perplexity, it required significantly more epochs, so LR=0.001 was preferred for its faster convergence.

Following this, we increased the batch size to 64 and 128 and scaled up the model capacity by raising the embedding dimension to 600 and 900, and the hidden size to 400 and 600. To compensate for the increased capacity, we applied dropout values of 0.3, 0.4, and 0.5. Based on the results, we further reduced the learning rate to 0.0005 and tested dropout rates of 0.5 and 0.7.

Finally, instead of increasing model width, we investigated increasing model depth. With the original dimensions of 200 (hidden) and 300 (embedding), we increased the number of layers (#L) from 1 to 3.

LSTM-AdamW (DropOut) Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	0.1	32	200	300	0.2	622.33
1	0.01	32	200	300	0.2	172.02
1	0.001	32	200	300	0.2	131.57
1	0.0001	32	200	300	0.2	131.29
1	0.001	64	200	300	0.2	130.90
1	0.001	128	200	300	0.2	128.96
1	0.001	128	400	600	0.3	121.10
1	0.001	128	600	900	0.3	117.44
1	0.001	128	400	600	0.4	120.14
1	0.001	128	600	900	0.4	116.70
1	0.001	128	400	600	0.5	119.23
1	0.001	128	600	900	0.5	116.56
1	5e-4	128	600	900	0.5	114.96
1	5e-4	128	600	900	0.7	121.04
2	5e-4	128	200	300	0.5	117.49
2	0.001	128	200	300	0.5	115.54
3	5e-4	128	200	300	0.5	118.84
3	0.001	128	200	300	0.5	121.13

Table 4: Results LSTM Networks with 2 Dropout Layers and AdamW optimizer

Test Result						
#L	LR	Batch	Hidden	Embed	DO	PPL
RNN						
1	1	128	200	300	/	157.00
LSTM						
1	2	32	200	300	/	139.17
LSTM - DO						
1	2	32	200	300	0.2	121.89
LSTM - DO - AdamW						
1	5e-4	128	600	900	0.5	105.69

Table 5: Results of 1A Testing

During testing, all models demonstrated improved perplexity, suggesting strong generalization and no signs of overfitting. The best-performing configuration employed the AdamW optimizer, even though it was the largest in terms of parameter count—typically a condition that increases the risk of overfitting. These results suggest that the AdamW optimizer contributes effectively to both stability and regularization.

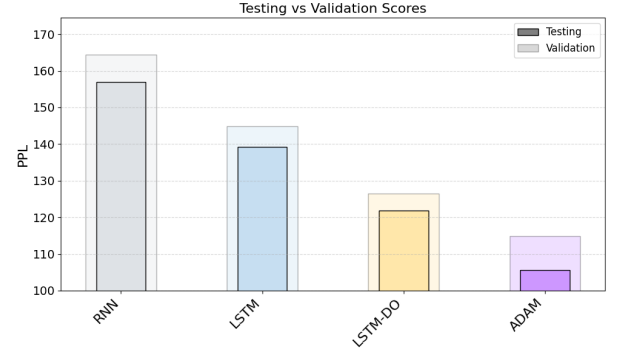


Figure 2: PPL difference between Validation and Testing

3.2. PART 1B

The point of start was the LSTM network, where the best configuration (LR:2, BS:32) reached a PPL in validation of 144.84.

WEIGHT TYING: This first regulation technique requires that the hidden dimension be equal to the embedding dimension. Using the same base configuration from the original LSTM network three different dimensions 100, 300, 500 were tested.

LSTM (WT) Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	2	32	100	100	/	131.87
1	2	32	300	300	/	119.32
1	2	32	500	500	/	120.79

Table 6: Results LSTM Networks with Weight Tying

VARIATIONAL DROPOUT: The second regularization technique was variational dropout, applied to both the input and output of the LSTM layers. I evaluated various dropout values 0.3, 0.5, 0.7 across dimensions of 300, 500, and 700.

LSTM (WT-VD) Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	2	32	300	300	0.3	100.33
1	2	32	300	300	0.5	95.98
1	2	32	300	300	0.7	106.7
1	2	32	500	500	0.3	100.56
1	2	32	500	500	0.5	92.96
1	2	32	500	500	0.7	92.39
1	2	32	700	700	0.3	104.29
1	2	32	700	700	0.5	94.68
1	2	32	700	700	0.7	89.27

Table 7: Results of LSTM Networks with Weight Tying and Variational Dropout

NT-AvSGD: The final improvement involved switching to Non-monotonically Triggered Averaged SGD (NT-AvSGD) as the optimization strategy. Given the extensive hyperparameter exploration in the previous stages, we limited this step to evaluating a higher dropout rate of 0.8, and a deeper architecture (2 layers) with hidden/embedding size of 450.

LSTM NT-AvSGD (WT-VD) Network						
#L	LR	Batch	Hidden	Embed	DO	PPL
1	3	32	700	700	0.7	87.35
1	3	32	700	700	0.8	93.13
2	3	32	450	450	0.5	93.71
2	3	32	450	450	0.7	90.06

Table 8: Results of LSTM Networks with Non-monotonically Triggered AvSGD, Weight Tying and Variational Dropout

Test Result						
#L	LR	Batch	Hidden	Embed	DO	PPL
Weight Tying						
1	2	32	300	300	/	116.06
Variational Dropout						
1	2	32	700	700	0.7	86.89
NTAvSGD						
1	3	32	700	700	0.7	84.64

Table 9: Results of 1A Testing

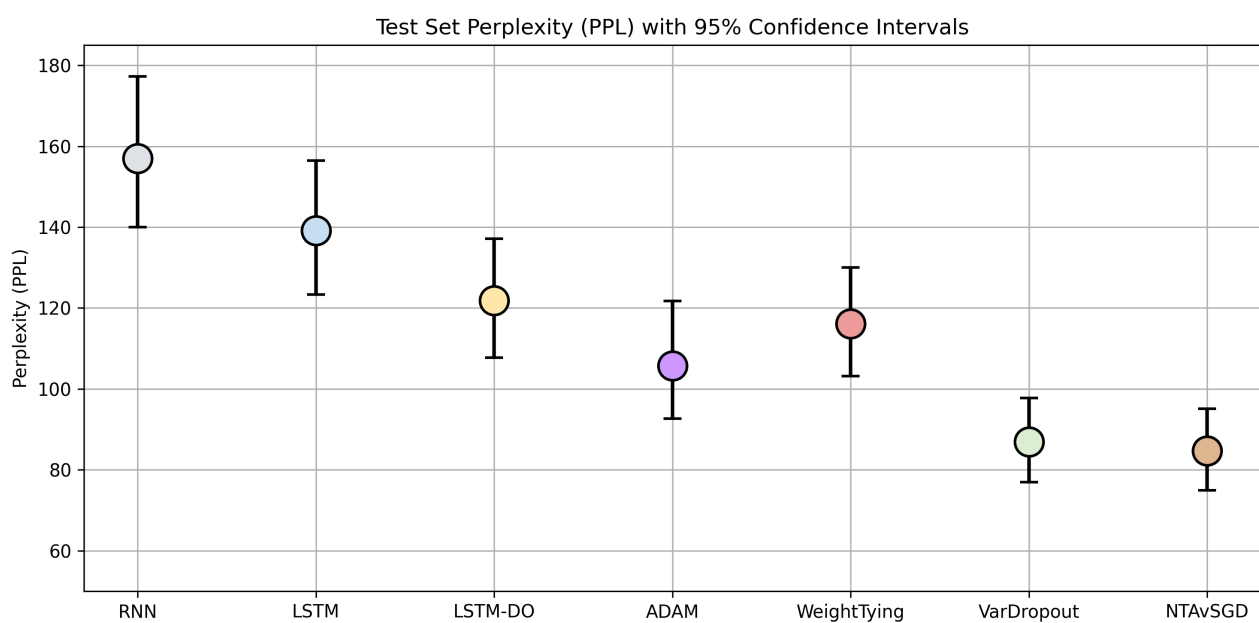


Figure 3: *PPL Plot of all the Tested models*

ID	Lable	LR	#L	Batch	Hidden	Embed	DO Emb	DO Out	Optimizer	Epoche	PPL	CIPPL
1	RNN	5	1	128	200	300	None	None	SGD	16	232.33	197.02-275.0
2	RNN	1	1	128	200	300	None	None	SGD	36	164.44	139.29-194.92
3	RNN	0.1	1	128	200	300	None	None	SGD	99	195.96	170.61-224.53
4	LSTM	0.5	1	16	200	300	None	None	SGD	18	146.94	130.69-165.58
5	LSTM	0.5	1	32	200	300	None	None	SGD	38	149.5	128.2-174.15
6	LSTM	0.5	1	64	200	300	None	None	SGD	46	155.47	134.48-179.73
7	LSTM	0.5	1	128	200	300	None	None	SGD	96	155.39	131.08-185.11
8	LSTM	1	1	16	200	300	None	None	SGD	12	145.96	129.72-164.79
9	LSTM	1	1	32	200	300	None	None	SGD	21	146.62	125.92-170.3
10	LSTM	1	1	64	200	300	None	None	SGD	32	150.28	129.19-175.45
11	LSTM	1	1	128	200	300	None	None	SGD	49	154.23	130.6-183.2
12	LSTM	2	1	16	200	300	None	None	SGD	9	145.42	128.75-164.63
13	LSTM	2	1	32	200	300	None	None	SGD	15	144.84	123.87-169.2
14	LSTM	2	1	64	200	300	None	None	SGD	19	148.38	127.49-173.02
15	LSTM	2	1	128	200	300	None	None	SGD	28	150.38	127.17-179.71
16	LSTM-DO	2	1	32	200	300	0.1	0.1	SGD	27	129.26	110.53-150.93
17	LSTM-DO	2	1	32	200	300	0.2	0.2	SGD	45	126.47	108.11-147.66
18	LSTM-DO	2	1	32	200	300	0.5	0.5	SGD	69	138.26	120.63-158.63
19	ADAM	0.1	1	32	200	300	0.2	0.2	AdamW	4	622.33	573.03-682.02
20	ADAM	0.01	1	32	200	300	0.2	0.2	AdamW	7	172.02	148.85-199.12
21	ADAM	0.001	1	32	200	300	0.2	0.2	AdamW	16	131.57	112.68-154.03
22	ADAM	0.0001	1	32	200	300	0.2	0.2	AdamW	99	131.29	113.21-152.99
23	ADAM	0.001	1	64	200	300	0.2	0.2	AdamW	22	130.9	111.64-154.42
24	ADAM	0.001	1	128	200	300	0.2	0.2	AdamW	33	128.96	108.36-158.78
25	ADAM	0.001	1	128	400	600	0.3	0.3	AdamW	21	121.1	101.31-148.52
26	ADAM	0.001	1	128	600	900	0.3	0.3	AdamW	16	117.44	97.76-145.33
27	ADAM	0.001	1	128	400	600	0.4	0.4	AdamW	27	120.14	100.38-148.55
28	ADAM	0.001	1	128	600	900	0.4	0.4	AdamW	19	116.7	97.2-144.8
29	ADAM	0.001	1	128	400	600	0.5	0.5	AdamW	38	119.23	99.25-148.3
30	ADAM	0.001	1	128	600	900	0.5	0.5	AdamW	23	116.56	97.09-143.46
31	ADAM	0.0005	1	128	600	900	0.5	0.5	AdamW	40	114.96	95.65-143.14
32	ADAM	0.0005	1	128	600	900	0.7	0.7	AdamW	83	121.04	101.08-150.66
33	ADAM	0.0005	2	128	200	300	0.5	0.5	AdamW	99	117.49	99.18-143.83
34	ADAM	0.0005	3	200	300	128	0.5	0.5	AdamW	86	118.84	100.4-144.74
35	ADAM	0.001	2	200	300	128	0.5	0.5	AdamW	77	115.54	96.45-143.96
36	ADAM	0.001	3	200	300	128	0.5	0.5	AdamW	60	121.13	101.66-149.35

Table 10: Complete Evaluation Results of part 1A

ID	Network Type	#L	Hidden	Embed	LR	Batch	DO	Optimizer	Epochs	PPL	95% CI PPL
1	WeightTying	1	100	100	2	32	None	SGD	27	131.87	114.62–151.95
2	WeightTying	1	300	300	2	32	None	SGD	17	119.32	102.36–138.95
3	WeightTying	1	500	500	2	32	None	SGD	14	120.79	103.31–141.05
4	VarDropout	1	300	300	2	32	0.3	SGD	36	100.33	85.72–117.42
5	VarDropout	1	300	300	2	32	0.5	SGD	60	95.98	82.33–111.85
6	VarDropout	1	300	300	2	32	0.7	SGD	80	106.70	92.72–122.96
7	VarDropout	1	500	500	2	32	0.3	SGD	26	100.56	85.74–117.93
8	VarDropout	1	500	500	2	32	0.5	SGD	42	92.96	79.25–109.04
9	VarDropout	1	500	500	2	32	0.7	SGD	77	92.39	79.17–107.68
10	VarDropout	1	700	700	2	32	0.3	SGD	20	104.29	88.96–122.26
11	VarDropout	1	700	700	2	32	0.5	SGD	32	94.68	80.56–111.19
12	VarDropout	1	700	700	2	32	0.7	SGD	60	89.27	76.25–104.51
13	NTAvSGD	1	700	700	3	32	0.7	NTAvSGD	59	87.35	74.52–102.41
14	NTAvSGD	1	700	700	3	32	0.8	NTAvSGD	91	93.13	79.91–108.46
15	NTAvSGD	2	450	450	3	32	0.7	NTAvSGD	28	93.71	79.86–109.83
14	NTAvSGD	2	450	450	3	32	0.8	NTAvSGD	48	90.06	76.92–105.15

Table 11: Complete Evaluation Results of part 1B

Lable	#L	LR	Batch	Hidden	Embed	DO Emb	DO Out	Optimizer	PPL	CI PPL
RNN										
RNN	1	1	128	200	300	/	/	SGD	157.00	140.01-177.29
LSTM										
LSTM	1	2	32	200	300	/	/	SGD	139.17	123.29-156.41
LSTM - DO										
LSTM-DO	1	2	32	200	300	0.2	0.2	SGD	121.89	107.76-137.12
LSTM - DO - AdamW										
ADAM	1	5e-4	128	600	900	0.5	0.5	AdmaW	105.69	92.6-121.71

Table 12: Results of 1A Testing

Lable	#L	LR	Batch	Hidden	Embed	DO	Optimizer	PPL	CI PPL
Weight Tying									
WeightTying	1	2	32	300	300	/	SGD	116.06	103.18-130.04
Variational Dropout									
VarDropout	1	2	32	700	700	0.7	SGD	86.89	76.99-97.74
NTAvSGD									
NTAvSGD	1	3	32	700	700	0.7	NTAvSGD	84.64	74.95-95.13

Table 13: Results of 1B Testing